



A two-stage evolutionary algorithm for uncertain constrained multi-objective problems with interval-valued objective

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ABSTRACT

In real-world engineering applications, there exist numerous constrained multi-objective optimization problems (CMOPs) that exhibit uncertain characteristics. Among these CMOPs, problems involving interval uncertainty in the objective functions pose significant challenges in simultaneously achieving optimal interval performance and constraint satisfaction. To address these challenges, a novel two-stage evolutionary algorithm, termed TSICMOEA, is proposed to solve interval constrained multi-objective optimization problems (ICMOPs) with interval-valued objectives. In the first stage, the primary goal of evolution is to conduct a global search efficiently. To accomplish this, an enhanced convergence indicator and a two-level dominance ranking strategy are proposed. These improvements aim to increase environmental selection pressure and generate as many valuable elite individuals as possible for the subsequent stage. In the subsequent stage, an adaptive elite selection strategy is employed, consisting of two cooperative subprocesses. These subprocesses collaboratively accelerate the population's convergence toward the feasible region and improve the quality of feasible solutions. Compared to other representative algorithms on two distinct ICMOPs benchmark suits with interval-valued objectives, the proposed TS-ICMOEA demonstrates significantly superior results for balancing the feasibility, convergence, and diversity performance characteristics, especially in ICMOPs with high-dimensional objective.

1. Introduction

Many optimization problems with multiple objectives inevitably need to consider complex equality or inequality constraints, such as engineering design [1], financial investment [2], and traffic optimization [3,4], which can be formulated as constrained multi-objective optimization problems (CMOPs). In these applications, the complex constraints make the search space of the problem discrete, shrink and change, which brings great challenges to solving CMOPs. Under this background, the research of constrained multi-objective optimization algorithms (CMOEA) with constraint processing techniques offers a viable and effective solution for addressing CMOPs.

Most of the existing studies formulating the CMOPs are based on parameters that are accurately measured. However, due to the absence of environmental factors or system-specific information, the parameters affecting the system model often exhibit complex

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uncertainties, making it difficult to accurately represent the actual conditions of engineering problems [5–7]. To obtain reliable data analysis and model design, valuable uncertainties need to be emphasized. Generally, there are three approaches to deal with uncertain problems [8,9], namely the stochastic, fuzzy, and interval methods. Optimization problems addressed by stochastic or fuzzy methods involve uncertain parameters, which are typically predefined as random variables or fuzzy numbers. It is often challenging and costly to obtain sufficient uncertainty information in practical engineering problems [10]. Compared with the above two approaches, obtaining the possible range of uncertain parameters is comparatively simpler and necessitates considerably less uncertainty information. Consequently, numerous scholars have devoted their efforts to the advancement of uncertainty modeling techniques incorporating interval parameters, upon which corresponding interval optimization methodologies have been introduced [11,12]. These methodologies have proven to be effective and efficient for addressing uncertain optimization problems that cannot be adequately handled by stochastic or fuzzy programming [13].

When solving interval constrained multi-objective optimization problems (ICMOPs) in which the objective functions are influenced by uncertain interval parameters, several critical issues need to be considered [14]. On the one hand, the ICMOPs with inequality or equality constraints lead to infeasible regions in the whole search space. For the optimal solutions that are often scattered in the feasible regions, the disconnected and isolated feasible regions make the Pareto Front (PF) locate on the boundary of the constraint. Hence, it is challenging to search for the optimal solutions if they are surrounded by infeasible regions [15]. On the other hand, ICMOPs challenge the traditional CMOPs when uncertain parameters are involved [16,17]. The Pareto optimality heavily depends on the interval boundaries of uncertain factors. Moreover, within the search space, the complex positional relationship among interval individuals makes it difficult to compare [18,19]. As a result, traditional Pareto dominance methods and population evolution strategies cannot be directly employed in multi-objective optimization models with uncertain parameters [20,21]. In response, a number of dedicated studies have emerged to specifically tackle the challenges associated with ICMOPs.

Currently, two common research methods have been employed for addressing ICMOPs. The first method is based on indirect conversion ideas [22]. By introducing certain decision parameters into an evaluation model, ICMOPs can be transformed into a deterministic multi-objective or single objective constrained optimization problem with real number parameters [23]. Then the optimal solutions of these transformed deterministic models can be obtained using conventional CMOEAs. For instance, Liu et al. [24] presented a multi-objective interval robust optimization model with interval characteristics for an inertial switch's bistable mechanism. They used the linear weighting method to transform the model into a single-objective optimization model and employed the penalty function method to achieve constraint satisfaction. Similarly, Liu et al. [25] transformed multiple interval-valued objectives into a single-objective optimization problem to construct an unconstrained multi-objective optimization problem, and used an improved micro multi-objective genetic algorithm to solve. In another approach, Cheng et al. [26] converted interval objectives and constraints into deterministic ones by utilizing the interval ranking relation, thereby enabling their solution using common NSGA-II-based evolutionary algorithms. Although these indirect transformation methods are relatively easy to solve, they tend to introduce instability in key interval information, resulting in slow convergence and unreliable feasible solutions [27].

The second category for solving ICMOPs adopts the interval multi-objective evolutionary algorithms (IMOEAs) with appropriate constraint handling mechanisms [28,29]. Xu et al. [30] employed the penalty function approach to convert ICMOPs into unconstrained problems. They introduced a novel interval crowding angle in an existing IMOEa, named IP-ICA-MOEa, which significantly improved the distribution performance. Liu et al. [31] handled constraints by measuring the degree of constraint violation, effectively converting them into unconstrained multi-objective uncertain optimization problems. Then an improved IMOEa named DI- μ MOGA was designed to solve. Chen et al. [32] improved NSGA-II for solving ICMOPs, where a P-dominance relation and constrained tournament rule were defined to select feasible high-quality individuals from the population. Zhang et al. [33] introduced an interval dominance relation to design an interval multi-objective particle swarm optimization algorithm (IPMOPSO). The Big-M penalty method was adopted as a considered handling approach for dealing with ICMOPs. To address vehicle scheduling and bandwidth allocation in federated learning environments, Cai et al. [34] designed a vehicle scheduling and bandwidth allocation scheme for federated learning by formulating an interval-constrained many-objective optimization model. An indicator-driven multi-level evolutionary algorithm was subsequently introduced to efficiently explore the feasible Pareto front. Seghir et al. [35] formulated an ICMOP for the web service composition and employed the constraint dominance principle (CDP) in their interval multi-objective artificial bee colony algorithm to prefer feasible individuals. While some of these ICMOEAs simultaneously handle constraints and objective functions, leading to the instability of feasible solutions and slow convergence, others emphasize the priority of constraints, focusing more on finding feasible regions but potentially sacrificing objective optimization under uncertainty [36,37]. Consequently, designing an efficient ICMOEA for solving ICMOPs remains a highly challenging task.

As mentioned above, an efficient ICMOEA must not only ensure the feasibility of balancing the population but also the search performance of the solutions within the interval-valued objective search space. Motivated by these observations, a novel two-stage interval constrained multi-objective evolutionary algorithm, named TS-ICMOEA, is proposed to deal with ICMOPs. The main contributions of this study are as follows:

- The proposed TS-ICMOEA comprises a two-stage process: the first stage promotes convergence of interval objectives for global search, and the second emphasizes constraints to balance objectives and feasibility.
- To improve the selection pressure of the population for quick convergence, a two-level dominance ranking method is proposed in the first stage, which provides as much promising information as possible for the subsequent stage.
- In the second stage, an adaptive elite selection strategy with two cooperative sub-processes is designed. The two sub-processes cooperatively guide the population to quickly approximate the feasible areas and enhance the performance of solutions while maintaining the feasibility.

The rest of this paper is organized as follows. Section 2 introduces the related work. The details of the proposed TS-ICMOEA are given in Section 3. The experimental results and analyses are presented in Section 4. Finally, the conclusion and future work are summarized in Section 5.

2. Preliminaries

In this section, the definition of ICMOPs with interval-valued objective is first introduced in Section 2.1. And some related interval definitions and interval domination relations for interval objective values are described in Section 2.2.

2.1. Interval constrained multi-objective optimization problem

The problem considered in this study is an ICMOP with uncertainty interval-valued objectives. In general, the mathematical formulation of ICMOP with uncertainty interval-valued objectives is given as [16]:

$$\begin{aligned} \min \quad & F(x, c) = (f_1(x, c), f_2(x, c), \dots, f_Z(x, c))^T \\ \text{s.t.} \quad & g_j(x) \leq 0, j = 1, 2, \dots, m, \\ & h_j(x) = 0, j = m + 1, \dots, n, \\ & x = (x_1, \dots, x_q) \in R_q, \\ & c = (c_1, \dots, c_p) \in R_p, c_l = [\underline{c}_l, \overline{c}_l], l = 1, 2, \dots, p \end{aligned} \quad (1)$$

where $F(x, c)$ is the objective function while $f_Z(x, c)$ is the Z -th interval-valued objective function determined by an uncertain parameter c ; $x = (x_1, \dots, x_q)$ is the q -dimensional decision vector; $c = (c_1, \dots, c_p)$ is the p -dimensional uncertain interval vector; $\underline{c}_l$ and $\overline{c}_l$ denote the lower and upper boundaries of the interval-valued parameter; $g_j(x)$ is the j -th inequality constraint, and $h_j(x)$ denotes the j -th equality constraint.

Definition 1. Constraint violation degree: In the ICMOP, the constraint violation (CV) degree is used for evaluating the feasibility of the candidate solutions. For an individual x , the CV on the j -th constraint is expressed as:

$$CV_j(x) = \begin{cases} \max\{g_j(x), 0\}, 1 \leq j \leq m \\ \max\{|h_j(x)| - \epsilon, 0\}, m + 1 \leq j \leq n \end{cases} \quad (2)$$

where ϵ denotes the tolerance parameter of the equality constraint, which is generally set to 0.0001. The overall CV of individual x is a measure that unifies the computation of all constraint violations, which can be calculated as:

$$CV(x) = \sum_{j=1}^n CV_j(x) \quad (3)$$

Definition 2. Feasible and infeasible solution: When an individual x satisfies the $CV(x) = 0$ for both the inequality constraint and the equality constraint, it is classified as a feasible solution. Otherwise, if $CV(x) \neq 0$, indicating that it does not satisfy one or more constraints, it is classified as an infeasible solution.

2.2. Dominance relation and crowding distance based on interval definition

Given two decision variables, x_i and x_j of Eq. (1), their z -th interval-valued objectives are $[f_z(x_i, c), \overline{f_z(x_i, c)}]$ and $[f_z(x_j, c), \overline{f_z(x_j, c)}]$ respectively. The interval probability defined in [31] is shown as Definition 3. Then the dominance relations and crowding distance definition based on interval probability are given as follows.

Definition 3. Interval probability: For two interval-valued objectives $I = [f_z(x_i, c), \overline{f_z(x_i, c)}]$ and $J = [f_z(x_j, c), \overline{f_z(x_j, c)}]$, if given a minimal interval value between I and J as $R = [\underline{R}, \overline{R}]$, where $\underline{R}$ and $\overline{R}$ are the smallest and second smallest of $U = [f_z(x_i, c), \overline{f_z(x_i, c)}, f_z(x_j, c), \overline{f_z(x_j, c)}]$, respectively. Then the probability of I less than J is defined as Eq. (4):

$$P(I < J) = \frac{d(J, R)}{d(I, R) + d(J, R)} \quad (4)$$

where $d(\cdot, \cdot)$ indicates the distance between two interval numbers. The interval distance $d(J, R)$ can be calculated as Eq. (5):

$$d(J, R) = \sqrt{(f_z(x_j, c) - \underline{R})^2 + (\overline{f_z(x_j, c)} - \overline{R})^2} \quad (5)$$

Similarly,

$$d(I, R) = \sqrt{(f_z(x_i, c) - \underline{R})^2 + (\overline{f_z(x_i, c)} - \overline{R})^2} \quad (6)$$

Based on the interval definition, the interval probability is employed to compare two interval number, the rules can be seen as following:

- If $P(I < J) > 0.5$, then $I < J$;

- If $P(I < J) < 0.5$, then $I > J$;
- If $P(I < J) = 0.5$, then $I = J$.

Definition 4. Interval dominance relation: An optimal solution x_i is said to dominate the solution x_j , if and only if:

$$x_i <_p x_j = \begin{cases} \forall k, P(I < J) \geq 0.5 \\ \exists k, P(I < J) > 0.5 \end{cases} \quad (7)$$

where $k \in \{1, 2, \dots, z\}$. Then the interval number x_i dominates x_j with the interval probability, denoted as $x_i <_p x_j$. If neither $x_i <_p x_j$ nor $x_j <_p x_i$, x_i and x_j are incomparable based on interval probability, denoted as $x_i \parallel_p x_j$.

Definition 5. Interval crowding distance: According to the Deb's crowding distance [38], the definition of crowding distance for individuals with interval-valued objective is updated. Specifically, the interval crowding distance ICD_r of the r -th solution can be calculated as follows:

$$ICD_r = \begin{cases} \infty, & r \text{ is the boundary solution} \\ \sum_{z=1}^Z \frac{d(f_z^{r+1}, f_z^{r-1})}{d(f_z, \bar{f}_z)}, & r \text{ is not the boundary solution} \end{cases} \quad (8)$$

The Eq. (8) holds if and only if r is not the boundary solution. If r is the boundary solution, then $ICD_r = \infty$. where $d(\cdot, \cdot)$ represents the distance between two interval-valued objectives, which can be measured using Eqs. (5) and (6); f_z^{r+1} and f_z^{r-1} denote the z -th objective of the neighbor solutions of the r -th solution from the sorted set.

3. Proposed algorithm: TS-ICMOEA

To cross through the infeasible domain and identify feasible domains, while ensuring that the CMOEA effectively explores elite solutions within the interval-objective space, it is essential to leverage interval-specific characteristics and incorporate a suitable constraint-handling mechanism in the development of the CMOEA. Taking these considerations into account, this study proposes a novel ICMOEA, named TS-ICMOEA. By incorporating the interval characteristics and adopting a suitable constraint handling approach, the TS-ICMOEA has sufficient ability to effectively address the challenges posed by ICMOPs and achieve satisfactory optimization results. In this section, we will introduce the framework of TS-ICMOEA in Section 3.1. Subsequently, we will delve into the algorithm details for the two stages strategy in Sections 3.2 and 3.3. Finally, we analyze the time complexity of TS-ICMOEA in Section 3.4.

3.1. The framework of TS-ICMOEA

The main components of the TS-ICMOEA consist of population initialization, fitness calculation, mating selection, genetic manipulation, and adaptive two-stage environmental selection. The adjustable two-stage framework is where the utmost effectiveness of TS-ICMOEA is demonstrated. The implementation process of TS-ICMOEA is depicted in Fig. 1, where the part enclosed in the dashed area represents the designed two-stage environmental selection procedure.

TS-ICMOEA divides the evolutionary process into two distinct stages, each tailored to address different aspects of solving ICMOPs. The first stage of the TS-ICMOEA focuses on global search capability, with the population evolution guided by elite individuals with superior global convergence. It incorporates a novel two-level ranking criterion to enable fast searches for populations with high selection pressure. This stage focuses on searching broadly within the interval objective space, allowing the algorithm to gather valuable information about promising regions, even if many individuals are still infeasible. Once the switching condition of the population evolution stage is met, the population transitions to the next stage. The purpose of the second stage is to guide the infeasible population towards feasible solutions and search for better performing solutions within the feasible region. The population leverages the search information obtained in the previous stage to enable a fast-approaching search towards the constrained PF, increasing the likelihood of the population escaping local feasible optima. As the population gradually moves toward the feasible region, the TS-ICMOEA is more likely to yield solutions with improved feasibility and effectiveness. Overall, this adaptive two-stage strategy enhances the effectiveness of TS-ICMOEA in balancing exploration and exploitation. Throughout the evolutionary process, the transitional state between the two stages is adjusted based on the evolutionary process control factor μ , which is related to the iteration condition.

Algorithm 1 outlines the overall procedure of TS-ICMOEA. Initially, a population P_t with N individuals is generated (line 1), where $t \in \{1, 2, \dots, T_{max}\}$ denotes the current iteration, and T_{max} represents the maximum number of iterations. The TS-ICMOEA then enters the evolutionary process loop (lines 3–13). In each generation, the interval-valued objective functions of all individuals in P_t are calculated, and then the interval dominance relationship and interval crowding distance are employed during mating selection to identify elite individuals as parent sets from P_t (line 4). The offspring population P_t^* is subsequently generated using genetic operators including simulated binary crossover and polynomial mutation (line 5). Next, the P_t and P_t^* are merged as a union population Q_t (line 6). When the t is less than $\mu * T_{max}$, the population Q_t enters the first evolutionary stage by default and executes the corresponding elite selection strategy (lines 7–8). Once the t exceeds $\mu * T_{max}$, the first stage terminates, and the elite selection strategy of the second stage begins to perform (lines 9–10).

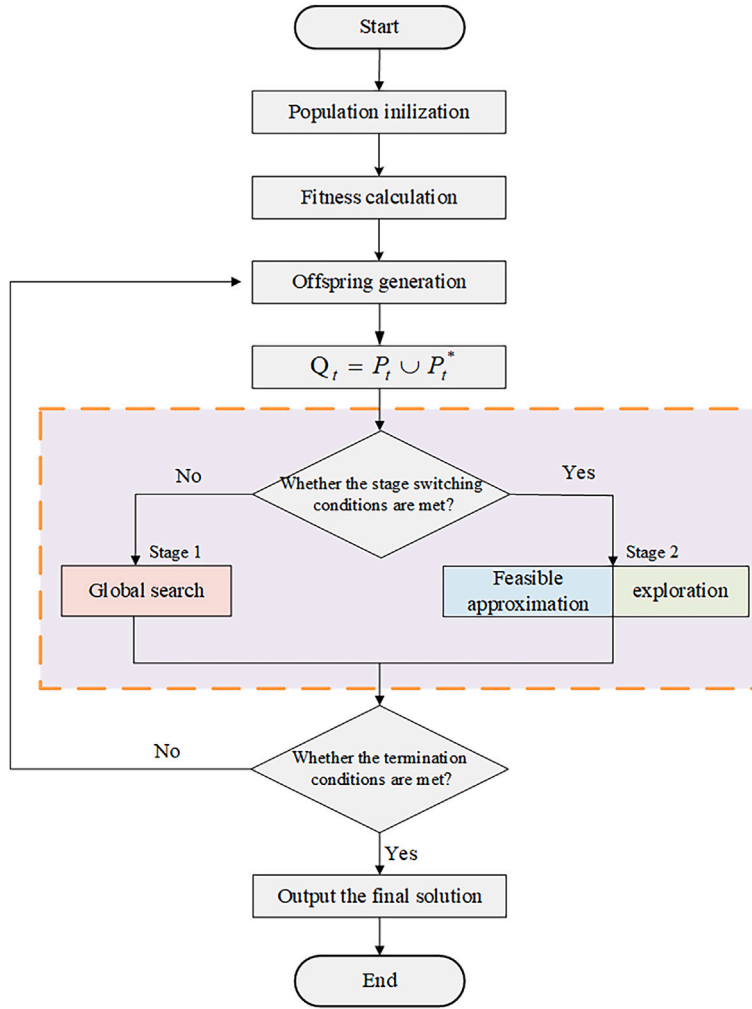


Fig. 1. The implementation process of TS-CIMOEA.

3.2. The first stage of global search

As previously described, the primary goal of the first stage is to carry out a global search effectively, enabling the transmission of valuable population information to the next phase of evolution. To explore wider promising regions and avoid population trapping in local feasible optima, some valuable infeasible individuals are utilized to strengthen the searchability and convergence [39,40]. In addition, as the number of objectives to be considered in ICMPs increases, relying solely on the interval dominance sorting method as the single ranking criterion may lead to insufficient selection pressure and poor search efficiency. To address this issue and better distinguish non-dominated individuals in a higher dimensional objective search space, a novel two-level balanced dominance sorting scheme is adopted, providing a stronger driving force to guide the population to converge rapidly.

For the proposed two-level balanced dominance sorting scheme, the first-level balance adopts the interval non-dominated sorting method between different objectives. The first-level balanced ranking model is formulated as Eq. (9):

$$\min([f_1(x), \overline{f_1(x)}], [f_2(x), \overline{f_2(x)}], \dots, [f_Z(x), \overline{f_Z(x)}]) \quad (9)$$

After the first-level ranking, individual x in the population is allocated a rank value. As the objective dimension increases, more individuals may end up with the same rank value. Using only a single ranking approach makes it difficult to dominate others in a higher dimensional objective space. In such cases, a additional ranking sorting can assist in distinguishing the elite individuals on the final non-dominated layer and guide the population towards convergence quickly. Considering the issue, a second-level ranking method is introduced, which is an unbiased bi-objective model that considers both convergence and feasibility of individuals. In the proposed model, the population optimizes the overall CV and a novel convergence indicator CI that assists the population in approaching the feasible boundary before entering the second phase. This second-level ranking method is formulated as Eq. (10):

$$\min(CI(x), CV(x)) \quad (10)$$

Algorithm 1 The framework of TS-ICMOEA.**Input:**Population size N , current evolutionary algebra t , maximum evolutionary algebra T_{max} ;**Output:**Final population $P_{T_{max}}$;

```

1: Initialize  $t = 0$ ;
2:  $P_t \leftarrow$  Population initialization( $N$ );
3: While  $t \leq T_{max}$  do
4:    $M_t \leftarrow$  Mating Selection( $P_t$ );
5:    $P_t^* \leftarrow$  Offspring( $M_t$ );
6:    $Q_t \leftarrow P_t \cup P_t^*$ ;
7:   If  $t \leq \mu * T_{max}$  then;
8:      $P_{t+1}^* \leftarrow$  Environment selection of first stage ( $Q_t$ ); //Refer to Algorithm 2
9:   Else
10:     $P_{t+1}^* \leftarrow$  Environment selection of second stage ( $Q_t$ ); //Refer to Algorithm 3
11:   End if
12:    $t++$ ;
13: End while
14: Return  $P_{T_{max}}$ 

```

Algorithm 2 The first stage of TS-ICMOEA.**Input:**Union population Q_t , Population size N **Output:**Next generation population P_{t+1}

```

1:  $k = 1, i = 1, L \leftarrow \emptyset, P_{t+1} \leftarrow \emptyset$ ;
2: Use interval nondominated sorting to divide  $Q_t$  into  $F_1, F_2, \dots$ 
3: While  $|P_{t+1}| + |F_i| \leq N$  do
4:    $P_{t+1} \leftarrow P_{t+1} \cup F_i$ ;
5:    $i = i + 1$ ;
6: End while
7:  $L \leftarrow$  Store elite individuals in last level  $F_i$  for further consider;
8: for each  $x \in L$  do
9:    $CI(x) \leftarrow$  Convergence Indicator Evaluation;
10:   $CV(x) \leftarrow$  Constraint violation value Calculation;
11: End for
12: Use Pareto non-dominated sorting to divide  $L$  into  $l_1, l_2, \dots$  by layer based on the designed model formulated in Eq. (10);
13: while  $|P_{t+1}| + |l_k| \leq N$  do
14:    $P_{t+1} \leftarrow P_{t+1} \cup l_k$ ;
15:    $k = k + 1$ ;
16: End while
17:  $P_{t+1} \leftarrow P_{t+1} \cup l_k$ ;
18: Truncation  $|P_{t+1}| + |l_k| - N$  individuals according to interval crowding distance;
19: End for
20: Return  $P_{t+1}$ 

```

where the CV value of individuals is measured as Eq. (3) and the convergence indicator CI is defined as the overall interval distance between the individuals x and the globally optimal ideal individual γ in the population, which is defined as follows:

$$CI = \sum_{z=1}^Z d([\underline{f_z(x)}, \overline{f_z(x)}], [\underline{f_z(\gamma)}, \overline{f_z(\gamma)}]) \quad (11)$$

where $[\underline{f_z(\gamma)}, \overline{f_z(\gamma)}]$ is the global minimal interval on the z -th objective, Similar to the interval variable K in Definition 3, $\underline{f_z(\gamma)}$ and $\overline{f_z(\gamma)}$ are the smallest and second smallest in the set composed of all individual interval objectives, respectively.

The second-level ranking method involves the ICMOP in the unconstrained bi-objective optimization problem. By performing this conversion, the population can effectively search in this 2-dimensional objective space. It allows for the application of the classical Pareto non-dominated sorting method to obtain additional rank value for the final level population, thereby enhancing both the searchability and convergence of the first stage in TS-ICMOEA.

Algorithm 3 The second stage of TS-ICMOEA.

Input:
Union population Q_t

Output:
Next generation population P_{t+1} , Population size N

```

1:  $P_{t+1} \leftarrow \emptyset$ ;
2: For each  $x \in Q_t$  do
3:   If  $CV(x) = 0$  then
4:      $SF \leftarrow SF \cup x$ ;
5:   Else
6:      $SI \leftarrow SI \cup x$ ;
7:   End if
8: End for;
9: if  $\varphi < 0.5$  do
10:   $P_{t+1} \leftarrow P_{t+1} \cup SF$ ;
11:  if  $|SF| \leq N$ 
12:     $CF = \text{sort}\{CV(SI)\}$ ;
13:     $P_{t+1} \leftarrow CF\{1 : N - |SF|\}$ ;
14:  End if
15: else
16:   $P_{t+1} \leftarrow \text{Select the top } N \text{ solutions from } SF$ ;
17: End if
18: Return  $P_{t+1}$ 

```

In cases where individuals within the same front cannot be further distinguished by the two-level dominance sorting method, the interval crowding distance is used as an auxiliary criterion. Individuals in the same layer are sorted based on their crowding distance values, and those with lower values are subsequently discarded. This strategy ensures a more refined selection pressure within the same dominance level, thereby preventing the population from being overwhelmed by excessive non-dominated individuals and ensuring the quality of elite selection. On the basis of the detailed procedure in the first stage of TS-ICMOEA, the pseudocode for the first stage of TS-ICMOEA is given in [Algorithm 2](#).

3.3. The second stage for feasible region approximation and exploration

Based on the population generated in the first stage, the assignment of the second stage is to further find the high-quality feasible population. Before considering the distribution and convergence of final solutions, it is essential to identify feasible solutions within the entire search space. To achieve this, two cooperative subprocesses are introduced: the feasible approximation subprocess and the exploration subprocess. The feasible approximation subprocess plays a crucial role in rapidly guiding the population towards the feasible regions, which helps the population quickly approach the feasible PF. In this process, well-performing individuals have priority in the next generation, and the CV degree of each individual is used as the auxiliary guiding criterion for elite individuals. The population tends to enter feasible areas to obtain more valuable information. The exploration subprocess focuses on continuously optimizing the population's convergence and distribution within the feasible domain. It aims to expand and converge the population along the feasible region, ensuring that all solutions are well distributed and exhibit good convergence properties. By cooperatively guiding the solutions, the two sub-processes jointly enable the population to quickly approximate the constrained PF, thereby enhancing the convergence performance of feasible solutions.

[Algorithm 3](#) describes the procedure for utilization in the second stage of TS-ICMOEA. Firstly, the input population Q_t is formed by combining the population P_t acquired in the previous stage and the offspring population P_t^* . The next step involves dividing the union population Q_t into two subpopulations based on the feasibility condition of the population. These subpopulations are referred to as the feasible subpopulation (SF) and infeasible subpopulations (SI). The execution of two sub-processes is determined by the feasible rate φ , which is calculated using [Eq. \(12\)](#). The feasible rate φ is measured by the ratio of the number of feasible individuals $|SF|$ to the total population size $|Q_t|$ and serves as a switching criterion for the following sub-processes.

$$\varphi = \frac{|SF|}{|Q_t|} \quad (12)$$

1) When the feasible rate φ is less than 0.5, it indicates that the number of feasible individuals $|SF|$ is less than the number of individuals to be selected N , and the individuals are predominantly distributed in the infeasible domain. In this case, it becomes crucial to identify more feasible solutions that dominate as many solutions as possible. Based on the consideration, all individuals in SF are considered to survive in the next generation P_{t+1} . And in order to quickly approach the boundary of the feasible domain, the remaining individuals in the infeasible subpopulation SI with lower CV are prioritized to enter P_{t+1} . By this way, the population will be led towards the feasible domain, allowing for the identification of more feasible solutions.

2) When the feasible rate φ exceeds 0.5, it shows that the majority of individuals in the population are generally within the feasible range. At this subprocess, there is sufficient candidate solution set space for our algorithm to select suitable elite individuals. Consequently, individuals need to be strictly sorted using an interval domination method based on the constrained dominance principle (CIDP), where the optimal feasible solutions have priority to be preserved in the next generation. By applying the CIDP, individuals are ranked according to their feasibility and interval domination relation, ensuring that feasible solutions dominate in terms of both feasibility and objective optimization. Additionally, the interval crowding distance method in Definition 5 is utilized to maintain diversity within the feasible set. This selection strategy considers the interval distance between individuals and their neighbors in the objective space, promoting a well-distributed population of feasible solutions. Ultimately, the first N best individuals, determined through the CIDP and interval crowding distance methods, are selected to form the next population P_{t+1} . By employing the feasibility-driven selection strategy in this subprocess, the population is guided towards maintaining feasibility, while also considering the optimization performance. The design of adaptive two subprocesses can explore more elite individuals with superior convergence performance within the feasible region.

3.4. Computational complexity of TS-ICMOEA

As described above, the TS-ICMOEA consists of five main parts, with the computational time complexity primarily dependent on the environmental selection in the two stages. Let N represent the population size, D denote the decision variables dimension, and M indicate the number of objectives. In the first phase, the complexity of the first-level balance scheme is $O(MN^2)$. Similarly, the complexity for solving the unbiased balance model in the second-level is also $O(MN^2)$. The complexity for crowding distance sorting is $O(N \log 2N)$. Hence, the worst-case complexity of environmental selection for the first stage is $O(MN^2)$. When the population enters the first subprocess in the second stage, the complexity of population sorting according to the CV degree is $O(MN^2)$ in the worst case. On the other hand, when the population enters the second subprocess, the complexity of the constrained interval non-dominated sorting method for feasible individuals is $O(MN^2)$. And the complexity for interval crowding distance sorting is $O(N \log 2N)$. Hence, the complexity of environmental selection for the second stage is $O(MN^2)$.

In summary, the computational complexity of TS-ICMOEA is $O(MN^2)$.

4. Experimental study

To evaluate the effectiveness of TS-ICMOEA for solving ICMOPs, a series of experimental studies are conducted on the PlatEMO platform [41]. We compared the proposed TS-ICMOEA with four representative approaches for solving ICMOPs, DI- μ MOGA [31], IMOE-DS [26], INSGA-II [32], and IP-ICA-MOE [30]. Since the IP-ICA-MOE is tailored for bi-objective optimization problems, its performance is evaluated only on benchmark problems involving two objectives. The comparison was performed on two sets of benchmark problems that involved interval-valued objectives and complex constraints. In Section 4.1, we provide detailed information about the benchmark suits used in the experiments. This includes a description of the problem instances and their interval objective characteristics. Additionally, in Section 4.2, we present the experimental settings, including the parameter configurations for each algorithm. Then the IIGD and feasible rate metrics used for evaluation are introduced in Section 4.3. The comparison results of TS-ICMOEA against the peer optimization methods on ICMOPs are presented in Section 4.4.

4.1. Benchmark suits

In our experiments, several test problems from LIRCOP, including LIRCOP1-LIRCOP4, and other DC-DTLZ benchmark problems are selected for validating the performance of TS-ICMOEA. These benchmark suites were selected because they consider constrained multi-objective decision problems that predominantly have an infeasible region within the target search space. This presents a significant challenge in balancing the search capability and feasibility for the algorithms.

Since this work focuses on evaluating the performance of TS-ICMOEA in solving ICMOPs, the aforementioned benchmark problems cannot be directly used in their original form. Therefore, it is necessary to convert traditional CMOPs into ICMOPs by introducing uncertainty into their objective functions. Following the approach taken in the literature on other interval optimization algorithms, we introduced disturbance factors in the LIRCOP and DC-DTLZ test functions to enable the extension of ICMEAs for benchmark optimization problems with interval-valued objectives [42]. Specifically, the disturbance factors η_z for the z -th objective used in each benchmark function are as follows:

$$\eta_z = \left| \sin(10 \cdot 2^{z-1} \pi \sum_d x_d) / 4 \right| \quad (13)$$

Therefore, we constructed a set of ICMOPs by transforming the objective functions of LIRCOP and DC-DTLZ into interval-based versions, which we refer to as InLIRCOP and InDC-DTLZ. These test suits pose a significant challenge for ICMEAs, as they require the algorithms to have the power to drive closer to the feasible domain of the constrained PF. The performance of ICMEAs in these test problems is of great significance, as it demonstrates their ability to handle ICMOPs effectively. Given the difficulty of crossing the infeasible region and converging towards the PF, achieving superior performance in these test problems showcases the capability of the ICMEAs.

4.2. Experimental settings

The common parameters in peer benchmark problems in the experiments are as Table 1. Moreover, to evaluate the searchability of our TS-ICMOEA on ICMOPs with high-dimensional objectives, we performed multiple assignments for the adjustable objective number

Table 1
Parameter settings of the test suits.

Test problem	Objective number (M)	Decision variable number (D)
InLIRCOMP1-InLIRCOMP4	2	30
InDC1-DTLZ1, InDC2-DTLZ1, InDC3-DTLZ1	2, 3, 4, 6, 8, 10, 15	6, 7, 8, 10, 12, 14, 19
InDC1-DTLZ3, InDC2-DTLZ3, InDC3-DTLZ3	2, 3, 4, 6, 8, 10, 15	11, 12, 13, 15, 17, 19, 24

Table 2
Common parameter settings of peer algorithms.

Variable name	Value
Population size (N)	100
Maximum number of iterations (T_{max})	1000
Crossover probability	1
Mutation probability	1/D

in the InDC-DTLZ benchmark suite. The parameter settings for the number of objectives and corresponding decision variables in the test sets are provided in Table 1. Some common parameter settings of peer algorithms are shown in Table 2.

Since the compared optimization approaches have different mechanisms for handling constraints, additional specific algorithm parameters need to be set. For the ICMOEAs using the penalty function approach, DI- μ MOGA, IMOEADS and IP-ICA-MOEa, set uniformly the penalty coefficient to 10,000 as stated in the literature [31]. Furthermore, for the solution method IMOEADS with the conversion idea, the interval-valued objective is transformed into deterministic ones by linear weighting of its midpoint and width. Different weight values indicate different emphasis on the converted optimization problem. Set the weight of midpoints to α , and the weight of widths to β . In the light of this information we set the weight with three cases: IMOEADS-1 ($\alpha=0.3$, $\beta=0.7$), IMOEADS-2 ($\alpha=0.5$, $\beta=0.5$), and IMOEADS-3 ($\alpha=0.7$, $\beta=0.3$). These three algorithms with different parameter settings are involved in the experiment. Furthermore, the parameter settings for all peer approaches are referred to from their respective original literature.

4.3. Performance metrics

To examine the effectiveness of TS-ICMOEA compared with other algorithms, we utilized two commonly used metrics to assess both convergence and diversity: Interval Inverted Generational Distance (IIGD) and Feasible Rate.

(1) IIGD metric

For the ICMOEA, the IIGD metric calculates the average Euclidean distance between the solutions of the constrained PF and the midpoint of the interval number obtained from the ICMOEA. The definition is as Eq. (14):

$$IIGD = \frac{\sqrt{\sum_{y^* \in PF^*} d(y^*, A)}}{PF^*} \quad (14)$$

where $d(y^*, A)$ represents the minimum Euclidean distance between y^* and the final solutions A . PF^* is the point set on the constrained PF and A denotes an approximate PF achieved by the ICMOEAs. A smaller IIGD result means better performance of a non-dominated set.

(2) Feasible rate metric

Additionally, the feasible rate metric calculated by Eq. (12) is also employed in our experiments, evaluating the feasibility of the comparison approach for solving ICMOPs.

4.4. Experimental analysis

4.4.1. Effects of different stage control factors μ

The stage control factor μ is an important parameter in TS-ICMOEA to adjust the evolutionary preference and strategy during the evolution. It can trigger a transition of two-stage environmental selection, allowing the population to be guided in a specified direction to approximate the true PF, which greatly influences the search efficiency and solution quality of ICMOEA. To study the performance effect of the stage control factor in TS-ICMOEA, we compared the TS-ICMOEA with different stage control factor values μ . The first variant is named TS-ICMOEA-group1 where μ is set to 0.3. The second variant is named TS-ICMOEA-group2 where μ is set to 0.7. The third and the last variant is our adopted TS-ICMOEA-group3 where μ is set to 0.5. Through all comparison algorithms performing 30 independent runs, the IIGD results on TS-ICMOEA variants with different factors on InLIRCOMP and InDC-DTLZ are shown in Table 3. Here, the Wilcoxon rank test results are provided in the last line. And the symbols ' $\pm$ ' indicate that the performance of TS-ICMOEA-group3 is statistically better than, worse than, or similar to the compared ICMOEAs.

Based on the IIGD results presented in Table 3, we can conclude from a detailed analysis that the proposed TS-ICMOEA with stage control factor $\mu=0.5$ achieves relatively better performance on the two sets of test suits. Specifically, our TS-ICMOEA-group3 has a similar or stronger advantage over TS-ICMOEA-group1 on all problems except InDC1-DTLZ1 with 3 objectives and InDC2-DTLZ3 with 10 objectives. The reason is that the TS-ICMOEA-group3 has sufficient time to search for high-quality individuals in the first stage, which greatly improves the search ability. Compared with the TS-ICMOEA-group2, our TS-ICMOEA-group3 is significantly better than TS-ICMOEA-group2 on 9 test problems, and slightly worse than TS-ICMOEA-group2 on the high-dimensional InDC1-DTLZ3 test

Table 3
The IIGD results of TS-ICMOEA with different control factors on ICMOPs.

Problem	M	TS-ICMOEA-group1	TS-ICMOEA-group2	TS-ICMOEA-group3
InLIRCOP1	2	2.5162e-1 (7.98e-2) =	3.0338e-1 (1.34e-1) =	2.4765e-1 (8.85e-2)
InLIRCOP2	2	2.2650e-1 (7.22e-2) =	1.8998e-1 (4.26e-2) =	2.1744e-1 (6.98e-2)
InLIRCOP3	2	2.8209e-1 (1.20e-1) =	2.7573e-1 (1.17e-1) =	2.9975e-1 (8.27e-2)
InLIRCOP4	2	2.9123e-1 (7.26e-2) =	2.5063e-1 (7.85e-2) =	2.6169e-1 (7.28e-2)
InDC1-DTLZ1	2	1.0409e-1 (1.22e-1) -	2.7666e-2 (4.62e-2) =	2.7650e-2 (4.63e-2)
	3	2.3055e-2 (2.79e-3) +	3.0983e-2 (7.97e-3) -	2.4861e-2 (3.19e-3)
	4	4.5207e-2 (4.64e-3) =	4.8426e-2 (7.30e-3) =	4.7102e-2 (6.02e-3)
	6	1.1713e-1 (4.04e-2) -	9.3424e-2 (1.07e-2) =	9.0521e-2 (8.47e-3)
	8	2.0408e-1 (7.87e-2) -	1.2733e-1 (2.40e-2) =	1.4768e-1 (4.55e-2)
	10	2.2518e-1 (5.84e-2) -	1.4322e-1 (2.13e-2) +	1.7524e-1 (5.18e-2)
	15	2.3186e-1 (5.88e-2) =	2.2154e-1 (4.09e-2) =	2.2217e-1 (4.82e-2)
InDC1-DTLZ3	2	5.8210e-1 (3.56e-1) =	4.2396e-1 (2.72e-1) =	4.8424e-1 (2.34e-1)
	3	3.5102e-1 (3.21e-1) =	1.8050e-1 (1.47e-1) =	2.1671e-1 (2.12e-1)
	4	4.2420e-1 (9.49e-2) -	3.8344e-1 (1.11e-1) =	3.5715e-1 (1.25e-1)
	6	7.6865e-1 (1.61e-1) -	5.7279e-1 (1.15e-1) +	6.6806e-1 (1.54e-1)
	8	2.3657e+0 (3.14e+0) -	8.4373e-1 (1.36e-1) +	1.0672e+0 (5.15e-1)
	10	6.9526e+0 (1.59e+1) -	9.8895e-1 (1.79e-1) +	4.8782e+0 (1.28e+1)
	15	1.0340e+1 (2.53e+1) =	1.2233e+0 (1.47e-1) =	1.3311e+1 (3.10e+1)
InDC2-DTLZ1	2	3.6135e-2 (5.53e-2) =	1.8247e-2 (1.33e-3) =	1.8194e-2 (2.01e-3)
	3	3.0917e-2 (1.94e-3) =	3.2351e-2 (3.59e-3) =	3.1082e-2 (3.07e-3)
	4	6.6322e-2 (4.75e-3) =	7.1392e-2 (9.53e-3) -	6.4933e-2 (3.14e-3)
	6	1.1550e-1 (6.60e-3) =	1.1981e-1 (1.92e-2) =	1.1328e-1 (6.92e-3)
	8	1.4718e-1 (2.02e-2) =	1.6639e-1 (2.30e-2) -	1.5156e-1 (2.78e-2)
	10	1.8175e-1 (4.11e-2) =	1.8125e-1 (2.92e-2) -	1.6195e-1 (1.84e-2)
	15	2.2899e-1 (4.85e-2) =	2.1840e-1 (2.37e-2) =	2.1252e-1 (3.14e-2)
InDC2-DTLZ3	2	5.5860e-1 (2.08e-4) =	3.6300e-1 (2.65e-1) =	3.9637e-1 (2.44e-1)
	3	5.9997e-1 (1.55e-2) -	2.1610e-1 (2.13e-1) =	3.6142e-1 (2.51e-1)
	4	6.5887e-1 (3.85e-2) =	4.5961e-1 (2.29e-1) +	6.3828e-1 (7.62e-2)
	6	8.0035e-1 (1.81e-2) =	7.2965e-1 (1.69e-1) =	7.9641e-1 (1.15e-1)
	8	9.3642e-1 (4.32e-2) =	9.8900e-1 (8.22e-2) -	9.2997e-1 (4.03e-2)
	10	1.0016e+0 (3.17e-2) +	1.0520e+0 (1.44e-1) -	1.0416e+0 (4.71e-2)
	15	1.2177e+0 (7.64e-2) =	1.3087e+0 (1.25e-1) -	1.2613e+0 (1.11e-1)
InDC3-DTLZ1	2	5.3097e-2 (8.77e-2) =	2.7669e-2 (4.63e-2) =	1.9215e-2 (3.91e-5)
	3	3.9418e-2 (3.90e-2) =	3.4053e-2 (3.60e-2) =	2.5767e-2 (2.60e-2)
	4	3.8290e-2 (3.83e-2) =	4.5801e-2 (4.59e-2) -	2.8104e-2 (2.07e-2)
	6	6.9240e-2 (7.62e-2) =	5.3566e-2 (2.60e-2) =	4.3279e-2 (1.98e-2)
InDC3-DTLZ3	2	1.1676e+0 (5.00e-1) -	7.1864e-1 (3.72e-1) =	8.2879e-1 (3.60e-1)
	3	1.1951e+0 (4.89e-1) -	4.4940e-1 (3.89e-1) =	5.8589e-1 (4.02e-1)
	4	1.0603e+0 (3.05e-1) -	6.6486e-1 (2.95e-1) =	7.4848e-1 (2.07e-1)
	6	1.2657e+0 (2.52e-1) =	1.1845e+0 (1.10e-1) =	1.1351e+0 (1.76e-1)
	8	1.3074e+0 (3.99e-1) -	1.1307e+0 (1.66e-1) -	1.0842e+0 (2.09e-1)
	10	1.7516e+0 (3.80e-1) -	1.4779e+0 (1.86e-1) =	1.5494e+0 (3.40e-1)
	15	6.0509e+0 (1.02e+1) -	2.0142e+0 (3.52e-1) =	2.2127e+0 (5.24e-1)
±/=		2/15/26	5/9/29	

problem with more than 6 objectives because the search time is less than TS-ICMOEA-group2. Adding μ can improve the global search performance, and it has a strong advantage in searching the feasible domain near PF by cross-domain infeasible region. For some InDC2-DTLZ1 and InDC2-DTLZ3 problems, our algorithm effectively explores the feasible domain and finds high-quality individuals in the feasible domain, thereby solving complex constraint problems more efficiently. Combined with the results of the Wilcoxon signed-rank, the TS-ICMOEA-group3 demonstrates a relatively obvious advantage compared with the algorithms under the other two parameter settings. To sum up, TS-ICMOEA can obtain considerable performance when $\mu = 0.5$. Accordingly, the stage control factor μ in the TS-ICMOEA is finally set to 0.5.

4.4.2. Performance comparison of TS-ICMOEA with other peer algorithms

To further compare the performance of the TS-ICMOEA with other peer algorithms, we conducted two sets of experiments. These experiments aimed to validate the effectiveness of TS-ICMOEA in solving different low and high-dimensional objective decision problems. The purpose of conducting experiments on problems with varying dimensions is to assess the scalability and robustness of the TS-ICMOEA algorithm.

In our first set of experiments, we tested the performance of the comparison algorithms in solving the InLIRCOP and InDC-DTLZ benchmark problems with low-dimensional objective spaces, specifically with 2 and 3 objectives. The results of these experiments are presented in Table 4. Table 4 shows the mean value and standard deviation of the IIGD obtained using the comparison ICMOEAs on the InLIRCOP and InDC-DTLZ problems. The term “NaN(NaN)” indicates that the obtained final solution does not include any feasible individuals. Additionally, Table 6 lists the average feasible rate of the final population obtained by the comparison algorithms over multiple independent runs. This metric indicates the percentage of feasible individuals in the final population obtained by each algorithm. These tables provide a comprehensive overview of the performance of the comparison ICMOEAs on low-dimensional

Table 4
The IIGD of different comparison algorithms on ICMOPs with 2 and 3 objectives.

Problem	M	DI- μ MOGA	IMOEA-DS-1	IMOEA-DS-2	IMOEA-DS-3	INSGA-II	IP-ICA-MOEA	TS-ICMOEA
InLIRCOP1	2	3.2800e-1 (3.31e-2) -	3.9378e-1 (0.00e+0) =	NaN (NaN)	NaN (NaN)	NaN (NaN)	3.5465e-1 (2.21e-2) -	2.4765e-1 (8.85e-2)
InLIRCOP2	2	2.8824e-1 (3.35e-2) -	3.2693e-1 (0.00e+0) =	NaN (NaN)	3.2492e-1 (8.19e-3) =	NaN (NaN)	3.1652e-1 (2.52e-2) -	2.1744e-1 (6.98e-2)
InLIRCOP3	2	3.2589e-1 (3.17e-2) =	NaN (NaN)	NaN (NaN)	3.7843e-1 (3.23e-2) =	NaN (NaN)	3.4861e-1 (3.25e-2) -	2.9975e-1 (8.27e-2)
InLIRCOP4	2	2.9746e-1 (3.35e-2) =	NaN (NaN)	3.4055e-1 (2.28e-2) =	3.3459e-1 (0.00e+0) =	NaN (NaN)	3.1838e-1 (1.72e-2) -	2.6169e-1 (7.28e-2)
InDC1-DTLZ1	2	4.6068e-2 (1.38e-1) -	2.3039e+2 (9.15e+1) -	1.7904e+2 (1.21e+2) -	1.2483e+2 (8.77e+1) -	3.8507e-2 (6.44e-2) -	1.9031e+0 (2.33e+0) -	2.7650e-2 (4.63e-2)
	3	3.1106e-2 (6.43e-3) -	1.7001e+2 (9.21e+1) -	1.7462e+2 (8.00e+1) -	9.0521e+1 (6.08e+1) -	7.8281e-2 (5.75e-2) -	-	2.4861e-2 (3.19e-3)
InDC1-DTLZ3	2	6.9279e-2 (5.12e-2) +	9.0728e+1 (2.39e+1) -	8.7086e+1 (2.51e+1) -	8.9399e+1 (2.41e+1) -	4.8799e-2 (1.46e-2) +	1.3292e+1 (2.67e+1) -	4.8424e-1 (2.34e-1)
	3	1.1369e-1 (4.82e-2) =	1.6937e+1 (3.46e+1) -	1.4621e+1 (3.05e+1) -	6.5898e+0 (2.40e+1) =	2.1405e-1 (1.35e-1) =	-	2.1671e-1 (2.12e-1)
InDC2-DTLZ1	2	2.0261e-1 (6.55e-3) -	3.3631e-2 (1.59e-2) -	2.6338e-2 (1.64e-2) -	1.9085e-2 (3.33e-3) =	4.3893e-2 (3.93e-2) -	4.8630e-1 (1.62e-1) -	1.8194e-2 (2.01e-3)
	3	1.8506e-1 (0.00e+0) =	7.5115e-2 (7.79e-2) -	4.6592e-2 (7.38e-2) =	3.1922e-2 (1.57e-2) -	7.0364e-2 (3.38e-2) -	-	3.1082e-2 (3.07e-3)
InDC2-DTLZ3	2	NaN (NaN)	7.4933e-1 (3.40e-1) -	4.6804e-1 (4.01e-1) =	4.2998e-1 (3.52e-1) =	1.8868e-1 (3.36e-1) +	5.5845e-1 (2.23e-4) =	3.9637e-1 (2.44e-1)
	3	NaN (NaN)	7.0336e-1 (2.13e-1) -	4.3068e-1 (3.34e-1) =	2.1944e-1 (2.74e-1) +	2.9752e-1 (2.79e-1) =	-	3.6142e-1 (2.51e-1)
InDC3-DTLZ1	2	1.3010e-1 (1.14e-1) -	2.4180e+2 (9.88e+1) -	1.9641e+2 (9.00e+1) -	1.3957e+2 (8.76e+1) -	2.5270e-2 (7.75e-3) -	9.2868e+0 (1.31e+1) -	1.9215e-2 (3.91e-5)
	3	9.5309e-2 (8.91e-2) -	2.0959e+2 (8.18e+1) -	2.0244e+2 (1.15e+2) -	1.0875e+2 (1.07e+2) -	1.0424e-1 (1.01e-1) -	-	2.5767e-2 (2.60e-2)
InDC3-DTLZ3	2	1.6271e+0 (3.77e-1) -	9.5120e+1 (2.68e+1) -	9.8188e+1 (2.34e+1) -	8.6511e+1 (3.03e+1) -	5.0993e-2 (1.38e-2) +	9.2524e+0 (1.11e+1) -	8.2879e-1 (3.60e-1)
	3	1.5356e+0 (6.44e-1) -	7.9546e+1 (2.41e+1) -	7.7636e+1 (2.50e+1) -	6.8937e+1 (2.39e+1) -	2.6394e-1 (3.40e-1) +	-	5.8589e-1 (4.02e-1)
$\pm/=$		1/9/4	0/12/2	0/9/4	1/8/6	4/6/2	0/9/1	

Table 5

The feasible rate of different comparison algorithms on ICMOPs with 2 and 3 objectives.

Problem	M	DI- μ MOGA	IMOEA-DS-1	IMOEA-DS-2	IMOEA-DS-3	INSGA-II	IP-ICA-MOEA	TS-ICMOEA
InLIRCMOP1	2	1.0000	0.03300	0	0	0	0.98733	1.0000
InLIRCMOP2	2	1.0000	0.03333	0	0.03667	0	0.95967	1.0000
InLIRCMOP3	2	1.0000	0	0	0.03300	0	0.97533	1.0000
InLIRCMOP4	2	1.0000	0	0.05667	0.03300	0	0.96633	1.0000
InDC1-DTLZ1	2	1.0000	1.0000	0.91700	0.95767	0.60533e	1.0000	1.0000
	3	1.0000	0.96133	0.97533	0.95500	0.48133	–	1.0000
InDC1-DTLZ3	2	1.0000	0.98767	0.99500	1.0000	0.60733	1.0000	1.0000
	3	1.0000	0.63400	0.57167	0.46700	0.80600	–	1.0000
InDC2-DTLZ1	2	0.26667	0.60567	0.77967	0.85267	1.0000	0.89967	1.0000
	3	0.03333	0.75400	0.89367	0.99500	0.99733	–	1.0000
InDC2-DTLZ3	2	0	0.16767	0.32733	0.33700	0.86433	0.06667	0.93333
	3	0	0.16367	0.38800	0.73467	0.94033	–	1.0000
InDC3-DTLZ1	2	1.0000	0.93767	0.90133	0.99600	0.44100	1.0000	1.0000
	3	1.0000	0.93567	0.92500	0.96867	0.21700	–	1.0000
InDC3-DTLZ3	2	1.0000	0.96100	0.99500	0.94667	0.51833	1.0000	1.0000
	3	1.0000	0.99433	0.92667	0.90533	0.29600	–	1.0000

objective problems, allowing for a comparison of their effectiveness in terms of convergence, diversity, and feasibility. By comparing the results of these experiments, we can gain a comprehensive understanding of how well the TS-ICMOEA algorithm performs in solving objective decision problems of varying objective dimensions.

InLIRCMOP is a representative bi-objective constrained interval optimization problem that features a large-scale infeasible region. The feasible PF of InLIRCMOP instances is often irregular or inconsistent with the unconstrained PF. It includes ICMOPs with various characteristics, some instances have highly narrow feasible regions that may even form curves. As shown in Table 4, TS-ICMOEA achieves the best performance on all InLIRCMOP. Moreover, the performance of our algorithm on InLIRCMOP1 and InLIRCMOP2 is the most prominent. The IIGD obtained by TS-ICMOEA, DI- μ MOGA, and IMOEA-DS-3 are statistically similar on InLIRCMOP3. The proposed TS-ICMOEA applies considerable pressure on the individuals to evolve towards the feasible PF. This allows the population to cross through the large-scale infeasible areas and search for the PF in InLIRCMOP. As shown in Table 5, our TS-ICMOEA can guarantee complete feasibility while maintaining considerable performance on the InLIRCMOP. These results highlight the effectiveness of TS-ICMOEA in handling the challenges posed by InLIRCMOP, including the large-scale infeasible region and irregular feasible PF. The algorithm demonstrates its ability to guide the search process towards the feasible PF while achieving competitive performance.

The InDC-DTLZ benchmark suite consists of problems with constraints on the decision space, categorized into three types: InDC1-DTLZ, InDC2-DTLZ and InDC3-DTLZ. These kinds of problems exhibit various combinations of constraint properties, such as continuity and discreteness, convexity and concavity. From Tables 4, we can observe that TS-ICMOEA has demonstrated significant advantages over other algorithms when solving the InDTLZ1 series problems with three types of constraints. By observing the performance results on the InDC2-DTLZ problems, it can be found that selection mechanisms based on feasibility preferences such as DI- μ MOGA, IMOEA-DS, and IP-ICA-MOEA when dealing with constrained optimization problems with discrete feasible regions, due to their weak global search ability, are prone to getting trapped in local feasible regions. Although the IIGD values of TS-ICMOEA are noticeably worse than those of DI- μ MOGA, INSGA-II, and IMOEA-DS-3 on InDTLZ3 with three types of constraints. One possible reason for this discrepancy is the strong selection pressure provided by the environment selection in the first stage of TS-ICMOEA. This pressure causes the population to quickly global search during the early stages of evolution, resulting in overshooting the feasible domain. Consequently, the population encounters difficulties in evolving towards the convex feasible PF in the second stage.

Additionally, when considering the feasible rate metric in Table 5, our algorithm demonstrates fully feasible performance on all InDC-DTLZ test problems, except for InDC2-DTLZ3 with 2 objectives. For the 2-objective InDC2-DTLZ3 test problem, this may be due to our algorithm's emphasis on obtaining considerable feasible solutions during the second stage, potentially sacrificing overall performance. These observations indicate that the performance of TS-ICMOEA can be influenced by the specific characteristics and combinations of constraints in the InDC-DTLZ problems. It underscores the significance of carefully considering selection preferences and negotiating a peace settlement between interval objective optimization and feasibility in the evolutionary process. By fine-tuning the algorithm's strategies and parameters, it may be possible to enhance its performance on challenging problems such as InDC2-DTLZ3. It is essential to note that the performance of TS-ICMOEA should be evaluated not only based on its feasibility but also its convergence and diversity. Achieving a balance among these factors is crucial for the algorithm's overall performance and effectiveness in solving complex ICMOPs.

According to the Wilcoxon-test results in Tables 4 and 5, it is evident that TS-ICMOEA consistently exhibited the best performance compared to DI- μ MOGA, IMOEA-DS-1, IMOEA-DS-2, IMOEA-DS-3, INSGA-II, and IP-ICA-MOEA on all two sets of ICMOPs with low-dimension objective. In terms of the IIGD metric, TS-ICMOEA consistently outperformed the other algorithms, with a significantly lower number of results worse than TS-ICMOEA compared to the number of results better than TS-ICMOEA. The number of results significantly worse than TS-ICMOEA ranged from 8 to 14, while the number of results significantly better than TS-ICMOEA ranged from 0 to 1, depending on the algorithm. Remarkably, there exist some cases where the results in the tables are marked as "NaN(NaN)", making it challenging to perform the Wilcoxon-test and leading to deviations in the statistical numbers. However, overall, the results suggest that TS-ICMOEA successfully converged and obtained superior solutions compared to the other algorithms on optimization

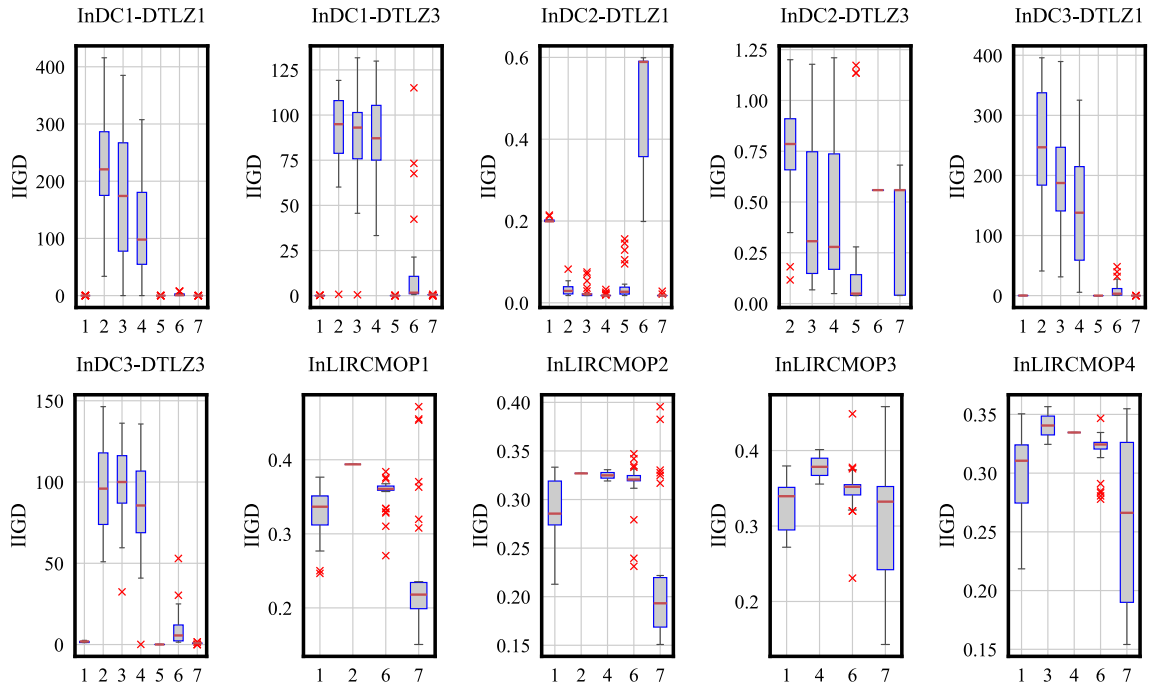


Fig. 2. Boxplots of IIGD values on 2-objective ICMOPs over 30 times: (1:DI- μ MOGA, 2:IMOEA-DS-1, 3:IMOEA-DS-2, 4:IMOEA-DS-3, 5:INSGA-II, 6:IP-ICA-MOEA, 7:TS-ICMOEA).

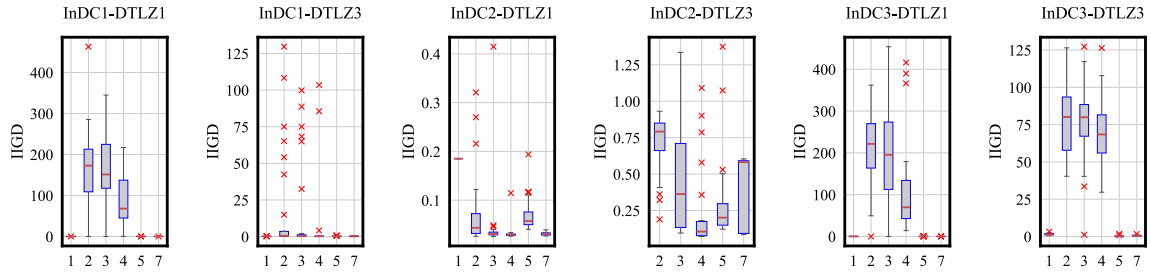


Fig. 3. Boxplots of IIGD values on 3-objective ICMOPs over 30 times: (1:DI- μ MOGA, 2:IMOEA-DS-1, 3:IMOEA-DS-2, 4:IMOEA-DS-3, 5:INSGA-II, 6:IP-ICA-MOEA, 7:TS-ICMOEA).

problems with 2 and 3 objectives, particularly in the InDC-DTLZ test suite with low-dimensional objectives. These findings highlight the effectiveness and superiority of TS-ICMOEA in tackling ICMOPs, particularly those with low-dimensional objectives and constraint properties.

The boxplots provide a visual representation of the algorithm's performance and demonstrate its superior search capabilities. Figs. 2 and 3 present the boxplots of the IIGD values for the five algorithms on InLIRC MOP and InDC-DTLZ test problems with 2 and 3 objectives. In these boxplots, the central mark of the box represents the median value, while the bottom and top lines of the box indicate the 1st quartile and 3rd quartile, respectively. From Fig. 2, it can be observed that the statistical average IIGD results obtained by TS-ICMOEA are significantly better than those of the other IC MOEAs on InLIRC MOP1 to InLIRC MOP4, InDC1-DTLZ1, InDC2-DTLZ1, and InDC3-DTLZ1 test problems. The boxplots for TS-ICMOEA exhibit the lowest median line and box length compared to the other IC MOEAs. This indicates that TS-ICMOEA demonstrates strong stability and convergence. While it is normal for TS-ICMOEA not to outperform other IC MOEAs on all test problems, the superior search performance and feasibility distribution advantages it displays on most of the test problems confirm the validity and feasibility of our algorithm. These findings highlight the effectiveness of TS-ICMOEA in solving the InLIRC MOP and InDC-DTLZ test problems with 2 and 3 objectives.

4.4.3. Performance results on ICMOPs more than 3 objectives

In this part of the experiments, we used the objective-adjustable InDC-DTLZ to evaluate the solution performance in high-dimensional objective optimization problems. Table 6 provides the statistical IIGD calculated by comparison IC MOEAs on the InDC-DTLZ with high-dimensional objectives. The IIGD metrics are employed to assess the convergence and coverage of the final non-dominated solutions. By analyzing the statistical results, we can compare the performance of the algorithms in terms of their ability to find high-quality solutions in the high-dimensional objective space.

Table 6

The IIGD results of different comparison algorithms on ICMOPs with 4–15 objectives.

Problem	M	DI- μ MOGA	IMOEA-DS-1	IMOEA-DS-2	IMOEA-DS-3	INSGA-II	TS-ICMOEA
InDC1-DTLZ1	4	7.4501e-2 (2.12e-2) -	1.6530e+2 (9.21e+1) -	1.2331e+2 (7.78e+1) -	4.8909e+1 (3.05e+1) -	1.6124e-1 (1.65e-1) -	4.7102e-2 (6.02e-3)
	6	1.1218e+0 (8.35e-1) -	1.1998e+2 (7.96e+1) -	9.0666e+1 (7.85e+1) -	5.0220e+1 (4.78e+1) -	6.2678e-1 (5.98e-1) -	9.0521e-2 (8.47e-3)
	8	2.6360e+0 (2.09e+0) -	1.0922e+2 (4.82e+1) -	9.6644e+1 (7.86e+1) -	6.0275e+1 (4.54e+1) -	8.1980e-1 (4.87e-1) -	1.4768e-1 (4.55e-2)
	10	2.2880e+0 (1.58e+0) -	1.1352e+2 (6.79e+1) -	6.8839e+1 (3.95e+1) -	4.1923e+1 (3.26e+1) -	1.5988e+1 (5.06e+1) -	1.7524e-1 (5.18e-2)
	15	3.0343e+0 (2.65e+0) -	1.2565e+2 (7.59e+1) -	9.4816e+1 (4.82e+1) -	7.1242e+1 (5.55e+1) -	4.1800e+0 (1.69e+1) -	2.2217e-1 (4.82e-2)
InDC1-DTLZ3	4	1.9086e+0 (8.86e-1) -	9.0574e+0 (1.88e+1) -	4.7875e+0 (1.00e+1) -	3.6613e+0 (2.14e+0) -	4.7363e-1 (2.16e-1) =	3.5715e-1 (1.25e-1)
	6	1.0497e+1 (3.07e+0) -	6.9643e+1 (1.80e+1) -	6.3061e+1 (1.65e+1) -	6.6953e+1 (1.88e+1) -	7.0945e+0 (2.58e+0) -	6.6806e-1 (1.54e-1)
	8	3.9188e+1 (2.53e+1) -	8.3505e+1 (1.78e+1) -	8.9045e+1 (2.71e+1) -	8.5812e+1 (2.02e+1) -	2.8221e+1 (1.61e+1) -	1.0672e+0 (5.15e-1)
	10	1.0620e+2 (2.85e+1) -	7.5218e+1 (2.57e+1) -	8.4402e+1 (1.98e+1) -	8.8637e+1 (2.32e+1) -	5.0164e+1 (2.51e+1) -	4.8782e+0 (1.28e+1)
	15	1.2922e+2 (2.42e+1) -	7.4325e+1 (2.28e+1) -	8.4323e+1 (1.84e+1) -	8.3695e+1 (1.67e+1) -	6.9629e+1 (3.57e+1) -	1.3311e+1 (3.10e+1)
InDC2-DTLZ1	4	1.9757e-1 (0.00e+0) =	2.8188e-1 (1.94e-1) -	2.1391e-1 (1.88e-1) -	1.1281e-1 (7.09e-2) -	1.1078e-1 (2.46e-2) -	6.4933e-2 (3.14e-3)
	6	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	2.4675e-1 (9.64e-2) -	1.1328e-1 (6.92e-3)
	8	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	3.0413e-1 (8.75e-2) -	1.5156e-1 (2.78e-2)
	10	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	3.6255e-1 (0.00e+0) =	1.6195e-1 (1.84e-2)
	15	2.2651e-1 (0.00e+0) =	NaN (NaN)	NaN (NaN)	NaN (NaN)	3.2575e-1 (0.00e+0) =	2.1252e-1 (3.14e-2)
InDC2-DTLZ3	4	6.6702e-1 (0.00e+0) =	7.2442e-1 (1.43e-1) =	7.5060e-1 (2.00e-1) -	7.5611e-1 (1.48e-1) -	7.3240e-1 (3.29e-1) -	6.3828e-1 (7.62e-2)
	6	8.2144e-1 (0.00e+0) =	NaN (NaN)	NaN (NaN)	NaN (NaN)	9.8230e-1 (1.27e-1) -	7.9641e-1 (1.15e-1)
	8	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	1.1467e+0 (2.81e-2) -	9.2997e-1 (4.03e-2)
	10	1.1605e+0 (0.00e+0) =	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	1.0416e+0 (4.71e-2)
	15	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	1.2613e+0 (1.11e-1)
InDC3-DTLZ1	4	1.9047e-1 (1.24e-1) -	1.9295e+2 (9.58e+1) -	1.5730e+2 (8.37e+1) -	1.1673e+2 (7.63e+1) -	4.2326e-1 (7.04e-1) -	2.8104e-2 (2.07e-2)
	6	3.0509e-1 (2.37e-1) -	1.8067e+2 (1.09e+2) -	1.1650e+2 (6.71e+1) -	7.9247e+1 (5.08e+1) -	1.0636e+2 (0.00e+0) =	4.3279e-2 (1.98e-2)
	8	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)
	10	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)
	15	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)	NaN (NaN)
InDC3-DTLZ3	4	2.4341e+0 (1.18e+0) -	7.4571e+1 (2.71e+1) -	6.8117e+1 (2.87e+1) -	5.2344e+1 (2.12e+1) -	1.4108e+0 (3.01e+0) =	7.4848e-1 (2.07e-1)
	6	1.3504e+1 (5.15e+0) -	4.4828e+1 (1.29e+1) -	5.3722e+1 (3.06e+1) -	4.3984e+1 (1.27e+1) -	3.2834e+0 (1.85e+0) -	1.1351e+0 (1.76e-1)
	8	3.8139e+1 (2.18e+1) -	7.4496e+1 (3.07e+1) -	8.4874e+1 (3.33e+1) -	9.1234e+1 (2.69e+1) -	1.8826e+1 (1.90e+1) -	1.0842e+0 (2.09e-1)
	10	6.6902e+1 (4.12e+1) -	8.6104e+1 (4.09e+1) -	1.0667e+2 (3.53e+1) -	9.8209e+1 (2.55e+1) -	6.2010e+1 (3.22e+1) -	1.5494e+0 (3.40e-1)
	15	7.5976e+1 (4.94e+1) -	1.0439e+2 (4.68e+1) -	9.5319e+1 (3.93e+1) -	1.0795e+2 (3.63e+1) -	NaN (NaN)	2.2127e+0 (5.24e-1)
$\pm/=$		0/17/5	0/18/1	0/19/0	0/19/0	0/19/5	

Table 7

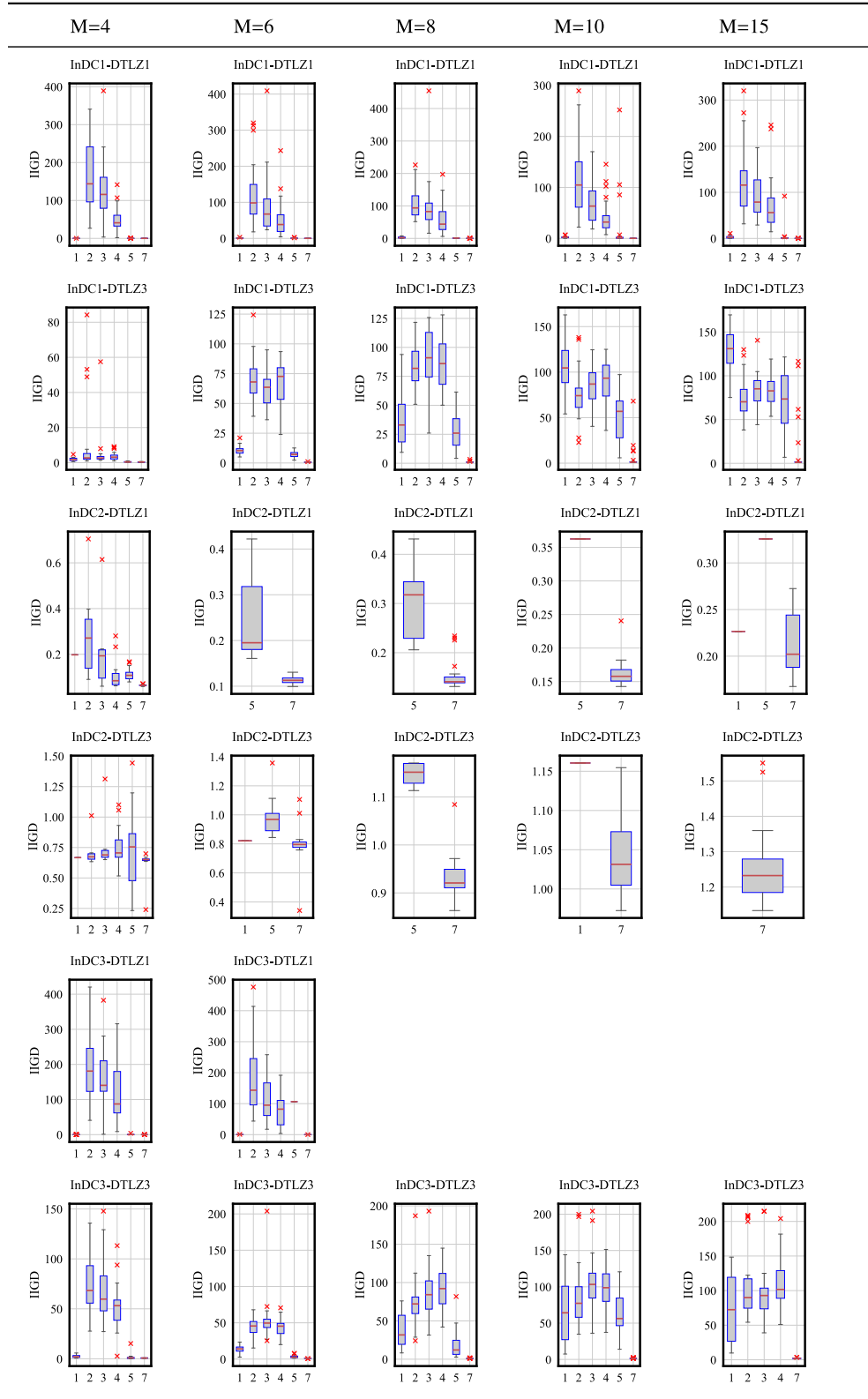
The feasible rate of different comparison algorithms on ICMOPs with 4–15 objectives.

Problem	M	DI- μ MOGA	IMOEA-DS-1	IMOEA-DS-2	IMOEA-DS-3	INSGA-II	TS-ICMOEA
InDC1-DTLZ1	4	1.0000	0.99967	0.97900	0.93200e	0.66733	1.0000
	6	1.0000	0.96833	0.99967	0.96767	0.60500	1.0000
	8	1.0000	1.0000	0.96700	0.99867	0.65500	1.0000
	10	1.0000	0.96967	1.0000	0.96533	0.53600	1.0000
	15	1.0000	1.0000	0.93900	0.96967	0.51933	1.0000
InDC1-DTLZ3	4	1.0000	0.81867	0.79733	0.78133	0.77967	1.0000
	6	1.0000	0.71833	0.69433	0.71000	0.65300	1.0000
	8	1.0000	0.70800	0.69867	0.66400	0.59900	1.0000
	10	1.0000	0.73433	0.71333	0.68467	0.55600	1.0000
	15	1.0000	0.83133	0.81800	0.81767	0.56433	1.0000
InDC2-DTLZ1	4	0.03333	0.86667	0.12600	0.28800	0.96133	1.0000
	6	0	0	0	0	0.16833	1.0000
	8	0	0	0	0	0.67667	1.0000
	10	0	0	0	0	0.00600	1.0000
	15	0.03333	0	0	0	0.00500	1.0000
InDC2-DTLZ3	4	0.03333	0.09067	0.18200	0.22700	0.30667	1.0000
	6	0.03333	0	0	0	0.21200	0.93333
	8	0	0	0	0	0.05700	0.86667
	10	0.03333	0	0	0	0	0.80000
	15	0	0	0	0	0	0.70000
InDC3-DTLZ1	4	1.0000	0.96733	0.88300	0.94000	0.07366	1.0000
	6	1.0000	0.93367	0.98567	0.91467	0.02667	1.0000
InDC3-DTLZ3	4	1.0000	0.94700	0.89933	0.84100	0.09767	1.0000
	6	1.0000	0.83800	0.90833	0.87533	0.07233	1.0000
	8	1.0000	0.27333	0.17467	0.11933	0.04233	1.0000
	10	1.0000	0.11700	0.02933	0.03233	0.00667	1.0000
	15	1.0000	0.04400	0.03567	0.02833	0	1.0000

It is evident from Table 6 that TS-ICMOEA consistently demonstrates the best IIGD among the algorithms on InDC-DTLZ instances with many objectives. The statistical results in the last row of the table confirm that no algorithm performs better than TS-ICMOEA for any of the test problems. As the number of objectives increases, the TS-ICMOEA exhibits more pronounced advantages compared to other algorithms. However, for InDC3-DTLZ1 with more than 8 objectives, the IIGD metrics cannot be measured. This is due to the complexity of the true constrained PF in the many-objective InDC3-DTLZ1 problem, which makes it challenging to generate within

Table 8

Boxplots of IIGD values on many-objective ICMOPs over 30 runs: (1: DI- μ MOGA, 2: IMOEA-DS-1, 3: IMOEA-DS-2, 4: IMOEA-DS-3, 5: INSGA-II, 6: IP-ICA-MOEA, 7: TS-ICMOEA).



the PlatEMO framework. It is indeed common for the selection pressure of existing peer algorithms to decrease when dealing with ICMOPs with many objective, which results in a weaker competitiveness compared to TS-ICMOEA. The powerful search capability of TS-ICMOEA in the early stages allows it to efficiently find well-performing solutions, even in small feasible regions and across large-scale infeasible regions, contributing to its superior performance on ICMOPs.

Furthermore, the feasible rate data provided in Table 7 highlights the superior feasibility performance of TS-ICMOEA on the InDC-DTLZ problems, with the exception of InDC2-DTLZ3. The challenging nature of the InDC2-DTLZ3 problem, characterized by a complex Pareto front and a large infeasible region, makes it difficult for all final individuals to converge to feasible solutions by jumping across the infeasible region. However, TS-ICMOEA still demonstrates excellent convergence and feasibility on this problem.

Based on the boxplots of the IIGD metrics for the comparison algorithms in Table 8, it is observed that TS-ICMOEA demonstrates superior performance compared to the other five ICMOEAs. Obviously, the box plotted for TS-ICMOEA is located in the lowest position and has the shortest box length on all test problems, which indicates it has the most centralized data distribution and the optimal average statistical value. The consistent superiority of TS-ICMOEA in these boxplots further confirms its effectiveness and competitiveness in solving the InDC-DTLZ problems with high-dimensional objective. Specifically, the figures for the InDC3-DTLZ1 are omitted when the number of objectives is set to 8, 10, and 15. Because all compared algorithms failed to obtain any feasible solutions under high-dimensional objective settings. Consequently, the IIGD values for all algorithms in these cases are NaN, rendering comparative plotting meaningless. This phenomenon is mainly due to the extreme difficulty of the InDC3-DTLZ1 problem under many-objective conditions. As the number of objectives increases, the feasible region becomes smaller and more complex to locate, especially under strict constraints and interval uncertainty.

In conclusion, the results of analyzing the above comparison experiments show that the proposed TS-ICMOEA demonstrates superiority over other optimization algorithms for solving ICMOPs. This is due to the novel two-stage strategy adopted to self-adaptively adjust the priority between objectives and constraints which maintains a trade-off between convergence and feasibility when dealing with different complex ICMOPs. Meanwhile, the elite individual selection pressure is enhanced by the two-level ranking strategy in the proposed TS-ICMOEA, resulting in the TS-ICMOEA exhibiting excellent search capability for handling ICMOPs, especially for ICMOPs with higher dimensional objective spaces.

5. Conclusion and future directions

In this study, a two-stage interval-constrained multi-objective evolutionary algorithm (TS-ICMOEA) is proposed to address interval-constrained multi-objective optimization problems (ICMOPs). The search process in TS-ICMOEA is divided into two stages, each employing distinct search directions and evolutionary strategies. In the first stage, the population performs a global convergence, and the utilization of promising elite individual information enhances the global search performance. To accomplish this, a comprehensive convergence indicator and a two-level domination ranking method are proposed to enhance selection pressure during the early evolutionary process. In the second stage, an adaptive elite selection strategy with two cooperative subprocesses is designed to evolve collaboratively toward the true PF. This strategy aims to further enhance the optimization process and improve the quality of the obtained solutions. Several sets of experiments demonstrate that our proposed TS-ICMOEA can improve the ability to search for high-quality feasible solutions of ICMOEAs while effectively achieving a balance between interval-valued objective optimization and constraint satisfaction.

While the two-stage strategy of TS-ICMOEA has significantly improved the convergence, feasibility and diversity of the ICMOEAs, it is essential to acknowledge that there are still some limitations and areas for further improvement. Future studies should focus on the application of TS-ICMOEA to other ICMOPs with complex interval constraints to evaluate its effectiveness in such scenarios. Additionally, the transmission factor of the proposed algorithm could be adjusted to optimize its performance further. By addressing these shortcomings and conducting additional studies, TS-ICMOEA can be further refined and enhanced to tackle a wider range of ICMOPs and improve its overall efficiency and effectiveness.

CRedit authorship contribution statement

Jie Wen: Writing – original draft, Supervision, Methodology, Formal analysis. **Qian Wang:** Visualization, Software, Data curation. **Zhihua Cui:** Writing – review & editing, Supervision, Project administration. **Jianghui Cai:** Writing – review & editing, Supervision. **Jinjun Chen:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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